8.4.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on nets?

Unit 8, Lesson 5: Surface Area of Rectangular Prisms and Pyramids

Benchmark

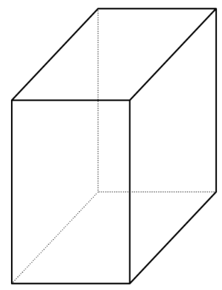
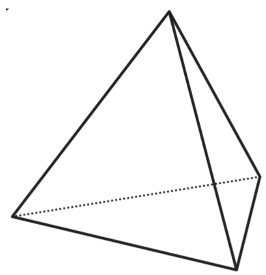
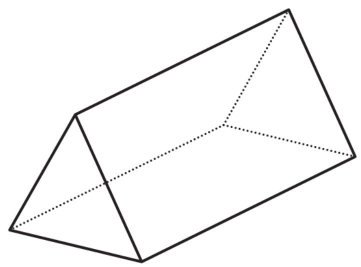
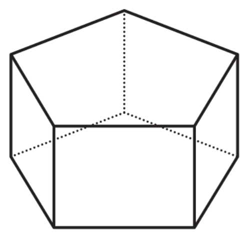
MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

Learning Targets

- ☐ I can find the surface area of a rectangular prism using a net or the figure.
- ☐ I can find the surface area of a rectangular pyramid using a net or the figure.

8.5.1 Warm-Up: Which One Doesn't Belong?

1. Four figures are shown.

A	B
	
C	**D**
	

   a. Which one doesn't belong?

   b. Justify your choice using mathematics.

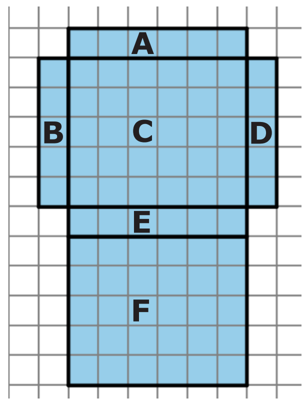
8.5.2 Guided Instruction: Surface Area of a Rectangular Prism

To find the surface area of a three-dimensional object, use the nets of a shape to find the area of each face.

> ***Surface area*** is the total area of the two-dimensional surfaces that make up a three-dimensional object

1. A net of a rectangular prism is shown below with each face labeled.

   a. Complete the table to find the area of each face. Each unit of the grid is 1 centimeter (cm).



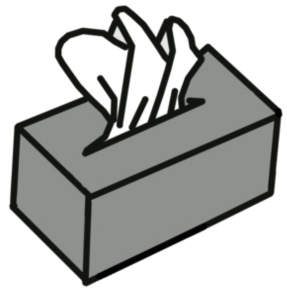
Face	Dimensions	Area
A		
B		
C		
D		
E		
F		

b. How can the net and the area of the six faces be used to find the surface area of the prism?

c. What is the surface area of this rectangular prism?

When calculating surface area, find the sum of the areas of each face of the figure. Use the figure's net to label the dimensions needed to calculate the area of each face.

2. A tissue box is 9 inches (in.) long, 5 in. wide, and 6 in. tall.



a. Sketch the net of the tissue box.

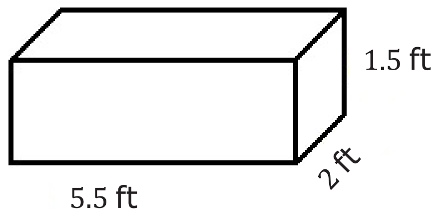
b. Label the dimensions of each face on the net.

c. Find the area of each face. Include units.

d. Add the area of the faces to find the surface area of the tissue box. Include units.

8.5.3 Collaboration: Analyzing Student Work

1. Jillian and Victor are finding the surface area, in square feet (ft^2), of the rectangular prism shown and each got the same answer; however, they solved the problem differently.

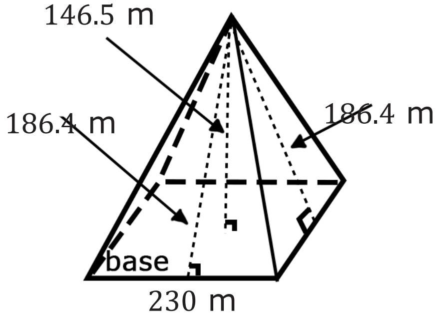


Jillian's Work	Victor's Work
Side 1: $5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$ Side 2: $5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$ Side 3: $5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$ Side 4: $5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$ Side 5: $2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$ Side 6: $2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$ Total Surface Area: 44.5 ft^2	2 sides: $2(5.5 \text{ ft} \times 1.5 \text{ ft}) = 16.5 \text{ ft}^2$ 2 sides: $2(5.5 \text{ ft} \times 2 \text{ ft}) = 22 \text{ ft}^2$ 2 sides: $2(2 \text{ ft} \times 1.5 \text{ ft}) = 6 \text{ ft}^2$ Total Surface Area: 44.5 ft^2

Discuss with your partner each student's strategy and how they are similar or different. Draw arrows to show connections between each student's work.

8.5.4 Guided Instruction: Surface Area of a Rectangular Pyramid

1. Around 2500 B.C. the Egyptians began building pyramids with intricate hallways and layouts to house the tombs of pharaohs and their families. One of the most famous is the Great Pyramid of Giza, also known as the Pyramid of Khufu, seen below. The Egyptians designed the pyramid to have a square base with approximate side lengths of 230 meters, a height of 146.5 meters, and a slant height of 186.4 meters.

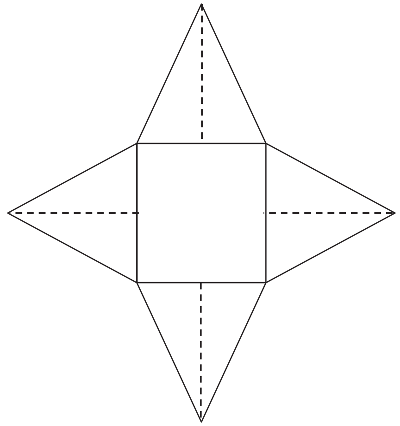


Looking at the model of the pyramid, why is there a measurement called height and a measurement called slant height?

The slant height of the pyramid represents the height of each triangular face.

When finding the surface area of a pyramid, use the **slant height** to find the area of the faces.

2. Using the information about the pyramid, complete the table to determine the surface area of the pyramid. Begin by labeling the sides of the net.



Side	Formula	Area
Rectangular base	$A = l \cdot w$	$A =$ ______ $\cdot$ ______ $A =$ ______
Triangular Face 1	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot$ ______ $\cdot$ ______ $A =$ ______
Triangular Face 2	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot$ ______ $\cdot$ ______ $A =$ ______
Triangular Face 3	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot$ ______ $\cdot$ ______ $A =$ ______
Triangular Face 4	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot$ ______ $\cdot$ ______ $A =$ ______
		Surface Area:

3. Complete the following statement.

To find the surface area of a rectangular pyramid you need the sum of the

○ height
○ area

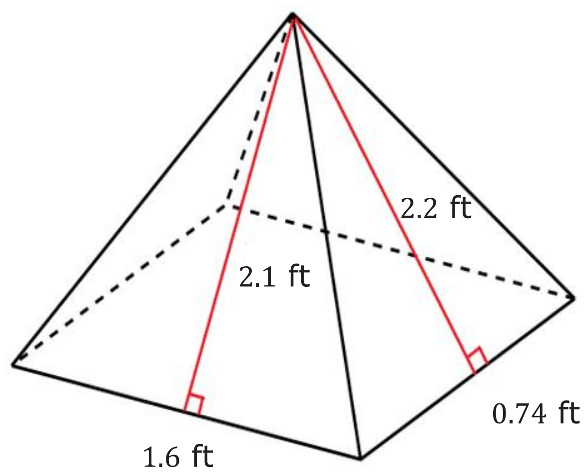
of the base plus the area of

○ one of
○ all

the other faces.

8.5.5 Collaboration: Surface Area of a Rectangular Pyramid

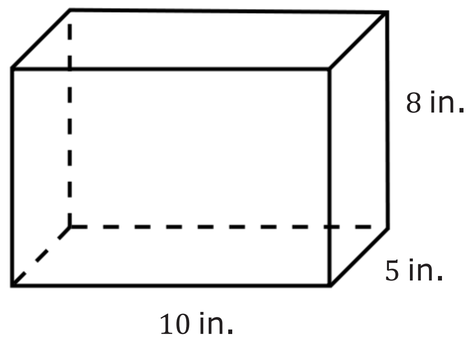
Papier-mache is a type of art that uses paper and glue. Jose is using the art of papier-mache to create a rectangular pyramid for his art project. To create his project, Jose needs to determine the amount of area that will be covered in newspaper. Below is the drawing Jose created to find the surface area.



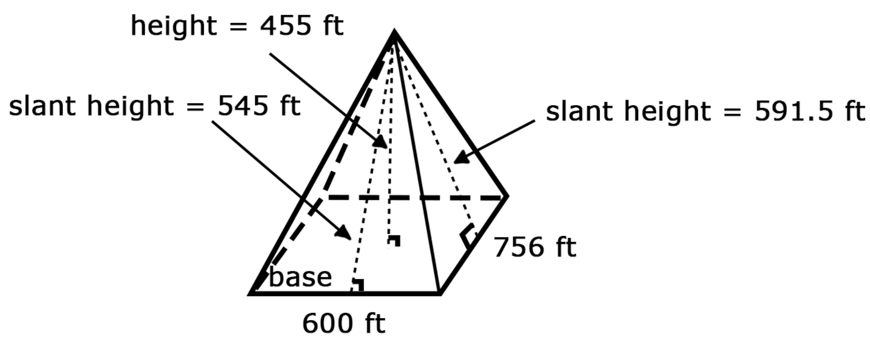
1. Work with your partner to find the surface area of the pyramid.

8.5.6 Your Turn!

1. Darsey wants to build trinket boxes made of wood to sell at the local craft festival. A drawing of her box is shown. How much wood does Darsey need to buy for each trinket box?



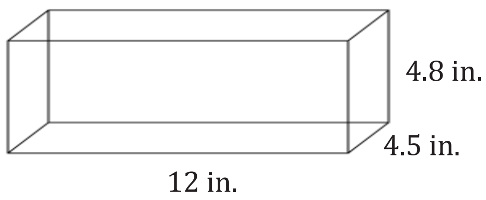
2. Find the surface area of a rectangular pyramid that has a rectangular base with a side length of 600 feet (ft), which is the base of the triangle with a slant height of 545 ft, and a side length of 756 ft, which is the base of the triangle with a slant height of 591.5 ft.



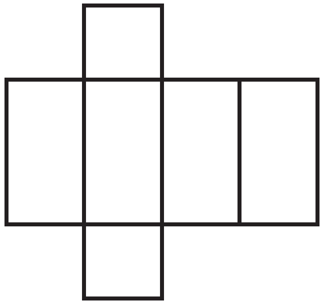
8.5.7 Wrap-Up: Ticket Out the Door

Ted is preparing a holiday display for the post office. He takes an unfolded box and uses it to determine the amount of wrapping paper needed to wrap the box.

The shipping box Ted uses for the display is shown.



1. Write the measurements on the net of the box.



2. Use the net to find the surface area of the shipping box. Include units.

Unit 8, Lesson 6: Finding Surface Area and Volume

Benchmarks

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.

Learning Targets

- ☐ I can find the surface area of right rectangular prisms and right rectangular pyramids.
- ☐ I can find the volume of right rectangular prisms.